Arithmetic-Heavy / QA Expected Paper

Disclaimer: This is a pattern-based predicted practice paper workflow output, not a leaked, exact, or official CAT paper.

Variant Blueprint

- Variant type: arithmetic_heavy
- Strategy: Lean into recurring QA priors.
- Target section mix: QA 22 from the frozen selected pool; VARC/DILR are outside the current candidate scope
- Expected overlap score: 0.865
- Risk score: 0.35
- Diversity score: 0.636

Questions

Q1. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Hard
- Score: 0.91

A shop buys 90 identical bags at Rs 640 each. It marks every bag at the same price. Of the 90 bags, 36 are sold at a 10% discount, 30 at a 20% discount, 12 at a 35% discount because of minor defects, and 12 remain unsold. If the shop still makes an overall profit of Rs 6600 on the entire purchase, what is the marked price of each bag?

A. Rs 900 B. Rs 960 C. Rs 1000 D. Rs 1080

Q2. QA - Arithmetic / Mixtures and Alligation

- Archetype/type: Mixture concentration problem
- Difficulty: Hard
- Score: 0.905

Solution A is 40% acid by volume and Solution B is 70% acid by volume. A chemist mixes a litres of Solution A with b litres of Solution B. After mixing, exactly 10 litres of pure water evaporates from the combined mixture. The remaining mixture has volume 40 litres and is 75% acid. Find the ratio $a : b$.

A. 1 : 1 B. 1 : 2 C. 2 : 3 D. 3 : 5

Q3. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.892

A value is first increased by p%, then decreased by 20%, and then again increased by p%. The final value is exactly 90% of what the value would have been after only a single p% increase from the original. If p is positive, find p.

Q4. QA - Arithmetic / Time and Work

- Archetype/type: Work-efficiency problem
- Difficulty: Medium
- Score: 0.882

A can finish a job alone in 30 days, while B can finish it alone in 20 days. A starts the job alone. B joins after 6 days and leaves as soon as only one-eighth of the job is left. A then completes the remaining work alone. In how many days is the job completed?

A. 17.25 days B. 17.85 days C. 18.25 days D. 18.75 days

Q5. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.873

The monthly incomes of P and Q are in the ratio 8:11, while their monthly expenditures are in the ratio 5:7. Q saves Rs 3000 more than P. After P invests 20% of his saving and Q invests 25% of his saving, their remaining savings are equal. What is Q's monthly income in rupees?

Q6. QA - Arithmetic / Averages

- Archetype/type: Weighted average problem
- Difficulty: Hard
- Score: 0.859

In a batch of 40 candidates, the average score is 69.6. The highest 5 candidates have an average score of 92, and the lowest 7 candidates have an average score of 48. The lowest 7 candidates are replaced by 7 new candidates. If the new average score of all 40 candidates becomes 74.5, find the average score of the 7 new candidates.

Q7. QA - Arithmetic / Time Speed Distance

- Archetype/type: Relative speed problem
- Difficulty: Hard
- Score: 0.83

Train A crosses a 150 m platform in 20 seconds. It overtakes a 180 m long train B, moving in the same direction, in 45 seconds. If train A is 250 m long, what is the speed of train B?

A. 32.4 km/h B. 36 km/h C. 37.6 km/h D. 40 km/h

Q8. QA - Arithmetic / Simple Interest Compound Interest

- Archetype/type: Interest-rate/time comparison problem
- Difficulty: Hard
- Score: 0.783

A sum of Rs. 64,000 is invested in Scheme P at compound interest of 12.5% per annum, compounded annually, for two years. The entire amount obtained at the end of two years is then reinvested in Scheme Q at simple interest of r% per annum for three years. The simple interest earned from Scheme Q is Rs. 7,300 more than the compound interest earned from Scheme P. Find r.

Q9. QA - Geometry / Triangles

- Archetype/type: Triangle similarity problem
- Difficulty: Medium-Hard
- Score: 0.865

In triangle ABC, point D lies on side AB and point E lies on side AC such that $DE \parallel BC$. The line through D parallel to AC meets BC at point F. If the area of triangle ADE is 16 and the area of quadrilateral $DECF$ is 40, find the area of triangle DBF.

A. 20 B. 24 C. 25 D. 36

Q10. QA - Algebra / Inequalities

- Archetype/type: Inequality optimization problem
- Difficulty: Hard
- Score: 0.855

For integer k with $0 \leq k \leq 15$, how many values of k make both roots of

$$x^2 + (k-3)x + (k-2) = 0$$

distinct negative real numbers?

A. 7 B. 8 C. 9 D. 12

Q11. QA - Algebra / Linear Equations

- Archetype/type: Integer feasibility under hidden divisibility constraint
- Difficulty: Hard
- Score: 0.854

A company has three departments X, Y, and Z. Every employee belongs to exactly one department. The number of employees in X is a positive multiple of 4. The number of employees in Y exceeds that in X by a positive multiple of

3. The number of employees in Z equals exactly half the sum of the employees in X and Y. If the total number of employees in the company is between 100 and 130, inclusive, how many distinct values can the total number of employees take?

Q12. QA - Algebra / Functions

- Archetype/type: Function value/domain problem
- Difficulty: Hard
- Score: 0.852

A function f defined on positive integers satisfies $f(1) = 1$ and

$$f(n) = n - f(f(n-1)) \quad \text{for } n \geq 2.$$

Find $f(20)$.

A. 10 B. 11 C. 12 D. 13

Q13. QA - Algebra / Logarithms

- Archetype/type: Base-change identity with algebraic cancellation
- Difficulty: Hard
- Score: 0.842

Let $a = \log_{12} 18$ and $b = \log_{24} 54$. ; What is the value of

$$\frac{1 - ab}{a - b} ?$$

A. 3 B. 5 C. 7 D. -5

Q14. QA - Geometry / Coordinate Geometry

- Archetype/type: Coordinate geometry line/slope/distance problem
- Difficulty: Hard
- Score: 0.839

In a parallelogram $ABCD$ with vertices taken in that order, $A = (1, 2)$ and $B = (7, 5)$. The vertex $C = (p, q)$ has integer coordinates and $p > 10$. The diagonals intersect at a point lying on the line $x + 2y = 16$, and the area of the parallelogram is 42. Find $p + q$.

Q15. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Medium
- Score: 0.908

A retailer buys 100 identical speakers at Rs 600 each and marks all of them at the same price. He sells 50 speakers at a discount of 10%, 30 speakers at a discount of 20%, and the remaining speakers, after minor carton damage, at a discount of 35%. If his overall profit is 9 1/3%, what is the marked price of each speaker?

A. Rs 760 B. Rs 780 C. Rs 800 D. Rs 840

Q16. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.887

In January, revenue from product X was exactly 40% of a firm's total revenue. In February, total revenue increased by 25%, but revenue from X decreased by 10% because only some contracts were renewed. By what percentage did revenue from all other products increase from January to February?

Q17. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Medium-Hard
- Score: 0.872

In an entrance test, 80% of candidates clear Section A. Of those who clear Section A, 75% clear Section B. Of those who clear both sections, 40% clear Section C. Among the candidates who do not clear all three sections, the candidates who failed only Section C (i.e., cleared A and B but failed C) form $\frac{9}{19}$ of that group. If the number of such candidates is 720, find the total number of candidates who appeared.

Q18. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.872

In an entrance test, 40% of the candidates are women. Exactly 30% of the men and 45% of the women qualify. Later, no qualified man withdraws, but some qualified women withdraw. After withdrawals, qualified men and qualified women are in the ratio 3:2. What percentage of the qualified women withdrew?

Q19. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.87

A fund is divided among A, B and C in the ratio 5:7:8. B gives Rs 22000 from his share to A, and then C gives one-fourth of his new comparison excess over A to B. After both transfers, A and B have equal amounts. What was C's original share in rupees?

Q20. QA - Arithmetic / Time and Work

- Archetype/type: Work-efficiency problem
- Difficulty: Hard
- Score: 0.862

A and B together can finish a job in 16 days, B and C together in 24 days, and A and C together in 18 days. A starts alone. After 4 days, C joins. After 3 more days, B also joins. How many days from the start are needed to finish the job?

A. 15.25 days B. 357/23 days C. 16 days D. 371/24 days

Q21. QA - Arithmetic / Time and Work

- Archetype/type: Work-efficiency problem
- Difficulty: Medium
- Score: 0.862

A can complete a job in 18 days and B can complete it in 27 days. A works from the start, while B joins after x days. The job is then completed, and their wages are divided in the ratio of the work actually done by A and B. If A receives Rs 5000 and B receives Rs 3000, then after how many days from the start did B join?

A. 1.125 days B. 1.5 days C. 2 days D. 2.25 days

Q22. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Hard
- Score: 0.853

A distributor buys 40 mixers at Rs 900 each and 80 more mixers at Rs 800 each. All 120 mixers are given the same marked price. During a festival, 30 are sold without discount, 50 are sold at a discount of 12%, and the remaining 40 are sold to a reseller at a discount of 25%. If the distributor's overall profit is 24.8%, what is the marked price of each mixer?

A. Rs 1120 B. Rs 1160 C. Rs 1200 D. Rs 1250

Answer Key

Q	Answer	Score
1	Rs 1000	0.91
2	(B) 1 : 2	0.905
3	12.5	0.892
4	17.85 days	0.882
5	825000	0.873
6	76	0.859
7	37.6 km/h	0.83
8	10	0.783
9	(C) 25	0.865
10	(B) 8	0.855
11	10	0.854
12	(C) 12	0.852
13	(B) 5	0.842
14	23	0.839
15	Rs 800	0.908
16	145/3	0.887
17	2000	0.872
18	100/3	0.872
19	144000	0.87
20	357/23 days	0.862
21	1.125 days	0.862
22	Rs 1200	0.853

Detailed Solutions

Q1

Step 1: Total cost of all 90 bags = 90 x 640 = Rs 57600. Step 2: Profit is calculated on the entire purchase, including the unsold bags, so required revenue = 57600 + 6600 = Rs 64200. Step 3: Let the marked price be M. Only 78 bags are sold, giving revenue 36 x 0.90M + 30 x 0.80M + 12 x 0.65M = 32.4M + 24M + 7.8M = 64.2M. Step 4: Therefore 64.2M = 64200, so M = Rs 1000. Step 5: Verification: revenue = 36 x 900 + 30 x 800 + 12 x 650 = 32400 + 24000 + 7800 = Rs 64200.

Q2

Step 1: After evaporation, the remaining volume is 40 litres and the acid concentration is 75%. Hence acid in the final mixture is

$$0.75 \times 40 = 30 \text{ litres.}$$

Step 2: Evaporation removes only water, not acid. Therefore acid before evaporation was also 30 litres. Step 3: Since 10 litres of water evaporated and final volume is 40 litres, the pre-evaporation mixed volume was

$$40 + 10 = 50 \text{ litres.}$$

So

$$a + b = 50.$$

Step 4: Acid balance before evaporation:

$$0.40a + 0.70b = 30.$$

Step 5: Substitute $a = 50 - b$:

$$0.40(50 - b) + 0.70b = 30,$$

$$20 - 0.40b + 0.70b = 30,$$

$$0.30b = 10 \Rightarrow b = \frac{100}{3}.$$

Then

$$a = 50 - \frac{100}{3} = \frac{50}{3}.$$

Step 6: Therefore

$$a : b = \frac{50}{3} : \frac{100}{3} = 1 : 2.$$

Q3

Step 1: Let the original value be 100. Step 2: After the three-stage change, the value is 100(1+p/100)(0.8)(1+p/100). Step 3: After only a single p% increase, the value would be 100(1+p/100). Step 4: The first value is 90% of the second, so 0.8(1+p/100)^2 = 0.9(1+p/100). Step 5: Since p is positive, divide by the common positive factor to get 0.8(1+p/100)=0.9, so p=12.5.

Q4

A's rate is 1/30 and B's rate is 1/20. In the first 6 days, A completes 6/30 = 1/5 of the job. B joins and they work together at rate 1/30 + 1/20 = 1/12 per day. B leaves when only 1/8 is left, so 7/8 of the job has been completed. The part done together is 7/8 - 1/5 = (35 - 8)/40 = 27/40. Time together = (27/40)/(1/12) = 81/10 = 8.1 days. The remaining 1/8 is done by A alone in (1/8)/(1/30) = 15/4 = 3.75 days. Total time = 6 + 8.1 + 3.75 = 17.85 days.

Q5

Step 1: Let incomes be 8x and 11x, and expenditures be 5y and 7y. Then P's saving is 8x - 5y and Q's saving is 11x - 7y. Step 2: Q saves Rs 3000 more, so 11x - 7y = 8x - 5y + 3000, giving 3x - 2y = 3000. Step 3: After investment, remaining savings are 80% of P's saving and 75% of Q's saving, so 0.8(8x - 5y)

= 0.75(11x - 7y). Since Q's saving exceeds P's by 3000, this is also 0.8(SQ - 3000) = 0.75SQ. Hence SQ = 48000. Step 4: Therefore 11x - 7y = 48000 and 3x - 2y = 3000. From the second equation, y = 1.5x - 1500. Step 5: Substitute: 11x - 7(1.5x - 1500) = 48000, so 0.5x + 10500 = 48000, x = 75000. Q's income = 11x = Rs 825000.

Q6

Step 1: Convert the original average into the original total:

$$40 \times 69.6 = 2784.$$

Step 2: The total score of the lowest 7 candidates is

$$7 \times 48 = 336.$$

Step 3: After replacing these 7 candidates, the total score of the unchanged 33 candidates is

$$2784 - 336 = 2448.$$

Step 4: The new average of all 40 candidates is 74.5, so the new total score is

$$40 \times 74.5 = 2980.$$

Step 5: Therefore, the total score of the 7 new candidates is

$$2980 - 2448 = 532.$$

Step 6: Hence their average score is

$$\frac{532}{7} = 76.$$

Q7

Step 1: While crossing the platform, Train A covers its own length plus platform length = 250 + 150 = 400 m in 20 seconds. So speed of A = 20 m/s. Step 2: While overtaking Train B in the same direction, Train A must cover the sum of both train lengths relative to B: 250 + 180 = 430 m. Step 3: Relative speed = 430/45 = 86/9 m/s. Step 4: Speed of B = 20 - 86/9 = 94/9 m/s. Step 5: Convert to km/h: (94/9) x 18/5 = 188/5 = 37.6 km/h.

Q8

Step 1: Since $12.5\% = \frac{1}{8}$, the two-year compound multiplier is

$$\left(1 + \frac{1}{8}\right)^2 = \left(\frac{9}{8}\right)^2 = \frac{81}{64}.$$

Step 2: Amount from Scheme P after two years is

$$64000 \times \frac{81}{64} = 81000.$$

So the compound interest earned in Scheme P is

$$81000 - 64000 = 17000.$$

Step 3: The simple interest earned in Scheme Q is Rs. 7,300 more than this, hence

$$SI_Q = 17000 + 7300 = 24300.$$

Step 4: In Scheme Q, the principal is not Rs. 64,000; it is the matured amount Rs. 81,000. Thus

$$24300 = \frac{81000 \cdot r \cdot 3}{100}.$$

Step 5: Solving,

$$r = \frac{24300 \cdot 100}{81000 \cdot 3} = 10.$$

Hence the required rate is 10% per annum.

Q9

Step 1: Let $\frac{AD}{AB} = k$. Since $DE \parallel BC$, triangle $ADE \sim ABC$ with linear ratio k, so $[ADE] = k^2 \cdot [ABC]$. Step 2: Since $DF \parallel AC$, triangle $DBF \sim ABC$ (same angle at B, with $DB/AB = 1-k$ and $BF/BC = 1-k$). Hence $[DBF] = (1-k)^2 \cdot [ABC]$. Step 3: Identify the hidden structure of $DECF$. Both $DE \parallel BC$ and $DF \parallel AC$. By the proportionality theorem, $DE = k \cdot BC$ and $FC = BC - BF = (1-(1-k)) \cdot BC = k \cdot BC$. So $DE = FC$ with $DE \parallel FC$, meaning $DECF$ is a *parallelogram*, not just a trapezoid. Step 4: Since the three regions $[ADE]$, $[DECF]$, $[DBF]$ partition $[ABC]$:

$$[ABC] = [ADE] + [DBF] + [DECF] = k^2T + (1-k)^2T + 2k(1-k)T = T.$$

So $[DECF] = 2k(1-k)\,T$. Step 5: Now $[DECF]^2 = 4k^2(1-k)^2T^2 = 4 \cdot [ADE] \cdot [DBF]$. Substituting:

$$40^2 = 4 \cdot 16 \cdot [DBF] \;\Rightarrow\; [DBF] = \frac{1600}{64} = 25.$$

Q10

Step 1: For both roots negative and *distinct* real:

- Discriminant > 0 (strict, for distinct): $(k-3)^2 - 4(k-2) > 0$.
- Sum of roots negative: $-(k-3) < 0$, i.e. $k > 3$.
- Product of roots positive: $k - 2 > 0$, i.e. $k > 2$.

Step 2: Simplify discriminant: $(k-3)^2 - 4(k-2) = k^2 - 6k + 9 - 4k + 8 = k^2 - 10k + 17 > 0$. Step 3: Roots of $k^2 - 10k + 17 = 0$: $k = \dfrac{10 \pm \sqrt{100-68}}{2} = \dfrac{10 \pm \sqrt{32}}{2} = 5 \pm 2\sqrt{2}$. Approximately $5 + 2\sqrt{2} \approx 7.83$ and $5 - 2\sqrt{2} \approx 2.17$. Step 4: So $k^2 - 10k + 17 > 0$ iff $k < 5 - 2\sqrt{2}$ or $k > 5 + 2\sqrt{2}$. Step 5: Combine with $k > 3$ and $k > 2$: intersection is $k > 5 + 2\sqrt{2} \approx 7.83$ (the branch $k < 5 - 2\sqrt{2} \approx 2.17$ is killed by $k > 3$). Step 6: Integer k with $7.83 < k \leq 15$: $k \in \{8, 9, 10, 11, 12, 13, 14, 15\}$. Count $= 8$. Step 7: Verify endpoints. $k = 8$: $x^2 + 5x + 6 = (x+2)(x+3) = 0$, roots $-2, -3$, both negative and distinct $\checkmark$. $k = 7$: $x^2 + 4x + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, complex roots $\times$ (good).

Q11

Step 1: Let the number of employees in X be $4a$ where $a \geq 1$, and let Y exceed X by $3b$ where $b \geq 1$. Then $|Y| = 4a + 3b$. Step 2: The number of employees in Z is

$$|Z| = \frac{|X| + |Y|}{2} = \frac{4a + 4a + 3b}{2} = \frac{8a + 3b}{2}.$$

For $|Z|$ to be a positive integer, $8a + 3b$ must be even. Since $8a$ is always even, $3b$ must be even, which forces b to be even. Write $b = 2k$ with $k \geq 1$. Step 3: With $b = 2k$, we get $|Z| = 4a + 3k$ and

$$T = |X| + |Y| + |Z| = 4a + (4a + 6k) + (4a + 3k) = 12a + 9k = 3(4a + 3k).$$

So T is always a multiple of 3. Step 4: Impose $100 \leq T \leq 130$:

$$100 \leq 3(4a + 3k) \leq 130 \implies 34 \leq 4a + 3k \leq 43.$$

Let $m = 4a + 3k$ where $a \geq 1$ and $k \geq 1$, so $m \geq 7$. For any integer $m \geq 7$, a representation $m = 4a + 3k$ with $a, k \geq 1$ exists: if $m \equiv 1 \pmod 3$, take $a = 1$; if $m \equiv 2 \pmod 3$, take $a = 2$; if $m \equiv 0 \pmod 3$, take $a = 3$; and solve for k accordingly. Each resulting k is positive for $m \geq 7$. Step 5: The integers m from 34 to 43 inclusive give $43 - 34 + 1 = 10$ values of m, hence 10 distinct totals $T \in \{102, 105, 108, 111, 114, 117, 120, 123, 126, 129\}$.

Q12

Step 1: Compute $f(n)$ step by step. $f(1) = 1.\backslash$ $f(2) = 2 - f(f(1)) = 2 - f(1) = 1.\backslash$ $f(3) = 3 - f(f(2)) = 3 - f(1) = 2.\backslash$ $f(4) = 4 - f(f(3)) = 4 - f(2) = 3.\backslash$ $f(5) = 5 - f(f(4)) = 5 - f(3) = 3.\backslash$ $f(6) = 6 - f(f(5)) = 6 - f(3) = 4.\backslash$ $f(7) = 7 - f(f(6)) = 7 - f(4) = 4.\backslash$ $f(8) = 8 - f(f(7)) = 8 - f(4) = 5$. Step 2: Continue. $f(9) = 9 - f(f(8)) = 9 - f(5) = 6.\backslash$ $f(10) = 10 - f(f(9)) = 10 - f(6) = 6.\backslash$ $f(11) = 11 - f(f(10)) = 11 - f(6) = 7.\backslash$ $f(12) = 12 - f(f(11)) = 12 - f(7) = 8$. Step 3: Continue. $f(13) = 13 - f(f(12)) = 13 - f(8) = 8.\backslash$ $f(14) = 14 - f(f(13)) = 14 - f(8) = 9.\backslash$ $f(15) = 15 - f(f(14)) = 15 - f(9) =$

9.\ $f(16) = 16 - f(f(15)) = 16 - f(9) = 10$. Step 4: Continue. $f(17) = 17 - f(f(16)) = 17 - f(10) = 11$.\ $f(18) = 18 - f(f(17)) = 18 - f(11) = 11$.\ $f(19) = 19 - f(f(18)) = 19 - f(11) = 12$.\ $f(20) = 20 - f(f(19)) = 20 - f(12) = 20 - 8 = 12$. Step 5: $f(20) = 12$.

Q13

Step 1: Set $p = \ln 2$ and $q = \ln 3$. Then

$$a = \frac{\ln 18}{\ln 12} = \frac{p+2q}{2p+q}, \qquad b = \frac{\ln 54}{\ln 24} = \frac{p+3q}{3p+q}.$$

Step 2: Compute $a - b$ over the common denominator $(2p+q)(3p+q)$:

$$a - b = \frac{(p+2q)(3p+q) - (p+3q)(2p+q)}{(2p+q)(3p+q)}.$$

Expanding the numerator:

$$(3p^2 + 7pq + 2q^2) - (2p^2 + 7pq + 3q^2) = p^2 - q^2 = (p-q)(p+q).$$

Step 3: Compute ab:

$$ab = \frac{(p+2q)(p+3q)}{(2p+q)(3p+q)} = \frac{p^2 + 5pq + 6q^2}{6p^2 + 5pq + q^2}.$$

Step 4: Compute $1 - ab$:

$$1 - ab = \frac{(6p^2 + 5pq + q^2) - (p^2 + 5pq + 6q^2)}{(2p+q)(3p+q)} = \frac{5p^2 - 5q^2}{(2p+q)(3p+q)} = \frac{5(p-q)(p+q)}{(2p+q)(3p+q)}.$$

Step 5: Form the ratio:

$$\frac{1-ab}{a-b} = \frac{5(p-q)(p+q)}{(2p+q)(3p+q)} \cdot \frac{(2p+q)(3p+q)}{(p-q)(p+q)} = 5.$$

Note that $p \neq q$ (since $\ln 2 \neq \ln 3$) and $p + q \neq 0$, so the cancellation is valid.

Q14

Step 1: In a parallelogram, the diagonals bisect each other. Hence the midpoint of AC is

$$\left(\frac{1+p}{2}, \frac{2+q}{2} \right).$$

Step 2: This point lies on $x + 2y = 16$, so

$$\frac{1+p}{2} + 2\left(\frac{2+q}{2}\right) = 16.$$

Step 3: Simplify:

$$\frac{1+p}{2} + 2 + q = 16.$$

Multiplying by 2,

$$1 + p + 4 + 2q = 32,$$

so

$$p + 2q = 27.$$

Step 4: The adjacent side vectors are

$$\overrightarrow{AB} = (6, 3), \qquad \overrightarrow{BC} = (p - 7, q - 5).$$

The area of the parallelogram is the absolute value of their determinant:

$$|6(q - 5) - 3(p - 7)| = 42.$$

Step 5: Substitute $p = 27 - 2q$:

$$|6q - 30 - 3(27 - 2q - 7)| = 42.$$

This becomes

$$|6q - 30 - 3(20 - 2q)| = 42,$$

or

$$|12q - 90| = 42.$$

Step 6: Therefore,

$$12q - 90 = 42 \quad \text{or} \quad 12q - 90 = -42.$$

So

$$q = 11 \quad \text{or} \quad q = 4.$$

Step 7: From $p + 2q = 27$, these give

$$(q, p) = (11, 5) \quad \text{or} \quad (4, 19).$$

The condition $p > 10$ selects

$$p = 19, \qquad q = 4.$$

Hence

$$p + q = 23.$$

Q15

Total cost price = 100 x 600 = Rs 60000. Let the common marked price be M. Revenue from the three groups is 50 x 0.90M + 30 x 0.80M + 20 x 0.65M = 45M + 24M + 13M = 82M. Overall profit is 9 1/3% = 28/3%, so required total selling price = 60000 x (1 + 7/75) = 60000 x 82/75 = Rs 65600. Hence 82M = 65600, so M = Rs 800. Verification: revenue = 50 x 720 + 30 x 640 + 20 x 520 = 36000 + 19200 + 10400 = Rs 65600.

Q16

Step 1: Let January total revenue be 100, so X revenue is 40 and other revenue is 60. Step 2: February total revenue is 125 after the 25% increase. Step 3: February X revenue is 90% of 40, which is 36. Step 4: Therefore February revenue from other products is 125 - 36 = 89. Step 5: The increase in other-products revenue is 29 on a base of 60, so the percentage increase is 2900/60 = 145/3%.

Q17

Step 1: Let total candidates = N. Clear A = $0.80N$. Clear A and B = $0.75 \times 0.80N = 0.60N$. Step 2: Clear all three = $0.40 \times 0.60N = 0.24N$. Hence candidates who do not clear all three = $N - 0.24N = 0.76N$. Step 3: Failed only C means cleared A and B but failed C, so this group is $0.60N - 0.24N = 0.36N$. Step 4: The given fraction is consistent because $0.36N/0.76N = 36/76 = 9/19$. Step 5: Given $0.36N = 720$, we get $N = 720/0.36 = 2000$.

Q18

Step 1: Let the total number of candidates be 100. Step 2: There are 60 men and 40 women; qualified men are 30% of 60 = 18. Step 3: Qualified women are 45% of 40 = 18 before withdrawals. Step 4: After withdrawals, men:women is 3:2, so if men are 18, remaining women are 12. Step 5: Women who withdrew = 18 - 12 = 6, which is 6/18 = 100/3% of qualified women.

Q19

Step 1: Let the original shares be 5x, 7x and 8x. After B gives Rs 22000 to A, A has 5x + 22000 and B has 7x - 22000. C still has 8x. Step 2: C's excess over A at this point is 8x - (5x + 22000) = 3x - 22000. One-fourth of this amount is transferred from C to B, so B receives (3x - 22000)/4. Step 3: Final A = 5x + 22000. Final B = 7x - 22000 + (3x - 22000)/4. Step 4: They are equal, so 5x + 22000 = 7x - 22000 + (3x - 22000)/4. Multiplying by 4 gives 20x + 88000 = 28x - 88000 + 3x - 22000 = 31x - 110000. Step 5: Hence 11x = 198000, so x = 18000. C's original share = 8x = Rs 144000.

Q20

Let the daily rates of A, B and C be a, b and c. We have a+b = 1/16, b+c = 1/24, and a+c = 1/18. Adding gives 2(a+b+c) = 1/16 + 1/24 + 1/18 = 9/144 + 6/144 + 8/144 = 23/144, so a+b+c = 23/288. Also a = [(a+b)+(a+c)-(b+c)]/2 = (1/16 + 1/18 - 1/24)/2 = (9+8-6)/144 divided by 2 = 11/288. Similarly c = [(b+c)+(a+c)-(a+b)]/2 = (6+8-9)/144 divided by 2 = 5/288. In the first 4 days A completes 4 x 11/288 = 44/288 = 11/72. In the next 3 days A+C complete 3 x (16/288) = 48/288 = 1/6. Work done in first 7 days = 11/72 +

1/6 = 23/72. Remaining work = 49/72. With all three working, time needed = (49/72)/(23/288) = 196/23 days. Total time = 7 + 196/23 = 357/23 days.

Q21

Since wages are proportional to work done, A's work : B's work = 5000 : 3000 = 5 : 3. Let the total completion time be T days. A works for T days and B works for T - x days. Their work amounts are T/18 and (T-x)/27. Hence (T/18)/((T-x)/27) = 5/3. This gives 3T/(2(T-x)) = 5/3, so 9T = 10T - 10x, hence T = 10x. Completion condition: T/18 + (T-x)/27 = 1. Substituting T = 10x gives 10x/18 + 9x/27 = 5x/9 + x/3 = 8x/9 = 1. Therefore x = 9/8 = 1.125 days.

Q22

Step 1: Total cost price = 40 x 900 + 80 x 800 = 36000 + 64000 = Rs 100000. Step 2: Let the marked price be M. The three realized selling-price groups contribute 30M + 50 x 0.88M + 40 x 0.75M = 30M + 44M + 30M = 104M. Step 3: A 24.8% overall profit means total revenue = 100000 x 1.248 = Rs 124800. Step 4: Hence 104M = 124800, so M = Rs 1200. Verification: revenue = 30 x 1200 + 50 x 1056 + 40 x 900 = 36000 + 52800 + 36000 = Rs 124800.

Evidence Lineage

Q	Candidate	Source Question IDs
1	CAT_QA_SPEC_01_ELITE_03	CAT_2017_S1_QA_Q12;CAT_2017_S2_QA_Q07
2	CAT_QA_SPEC_10_CORRECTED_104	CAT_2018_S1_QA_Q01;CAT_2018_S1_QA_Q14
3	CAT_QA_SPEC_06_ELITE2_03	CAT_2017_S1_QA_Q01;CAT_2017_S1_QA_Q05
4	CAT_QA_SPEC_02_CAND_01	CAT_2017_S1_QA_Q02;CAT_2017_S1_QA_Q15
5	CAT_QA_SPEC_03_ELITE_01	CAT_2017_S1_QA_Q13;CAT_2018_S1_QA_Q26
6	CAT_QA_SPEC_15_GPT55Thinking_AVGDM_01	CAT_2017_S1_QA_Q21;CAT_2017_S2_QA_Q26
7	CAT_QA_SPEC_04_ELITE_01	CAT_2017_S1_QA_Q04;CAT_2017_S2_QA_Q04
8	CAT_QA_SPEC_09_CORRECTED_101	CAT_2018_S1_QA_Q15;CAT_2019_S1_QA_Q11
9	CAT_QA_SPEC_07_CLAUDE_01	CAT_2017_S1_QA_Q16;CAT_2018_S1_QA_Q09
10	CAT_QA_SPEC_08_CLAUDE_07	CAT_2017_S1_QA_Q03;CAT_2017_S1_QA_Q26
11	CAT_QA_SPEC_12_GENSPARK_FINAL_01	CAT_2017_S1_QA_Q28;CAT_2018_S2_QA_Q29
12	CAT_QA_SPEC_13_CLAUDE_04	CAT_2018_S2_QA_Q30;CAT_2019_S1_QA_Q09
13	CAT_QA_SPEC_16_GENSPARK_FINAL_01	CAT_2017_S1_QA_Q23;CAT_2018_S1_QA_Q11
14	CAT_QA_SPEC_18_GPT55Thinking_COORDMENT_04	CAT_2017_S1_QA_Q24;CAT_2019_S1_QA_Q03
15	CAT_QA_SPEC_01_CAND_01	CAT_2017_S1_QA_Q08;CAT_2017_S1_QA_Q10
16	CAT_QA_SPEC_06_ELITE2_01	CAT_2017_S1_QA_Q01;CAT_2017_S1_QA_Q05
17	CAT_QA_SPEC_06_CLAUDE_06	CAT_2017_S2_QA_Q06;CAT_2017_S2_QA_Q10
18	CAT_QA_SPEC_06_ELITE2_02	CAT_2017_S1_QA_Q01;CAT_2017_S1_QA_Q05
19	CAT_QA_SPEC_03_ELITE_03	CAT_2018_S1_QA_Q26;CAT_2018_S2_QA_Q34
20	CAT_QA_SPEC_02_CAND_03	CAT_2017_S1_QA_Q34;CAT_2017_S2_QA_Q34

Q	Candidate	Source Question IDs
21	CAT_QA_SPEC_02_CAND_04	CAT_2017_S1_QA_Q15;CAT_2017_S2_QA_Q13
22	CAT_QA_SPEC_01_ELITE_01	CAT_2017_S1_QA_Q08;CAT_2017_S1_QA_Q10